Numerical Reproduction of Energies and angular momenta of periodic Schwarzschild geodesics

1. Overview

The paper studies periodic geodesics around a Schwarzschild black hole and their distribution in the (L, E)-parameter space. Periodic orbits are classified by the Levin–Perez-Giz taxonomy (z, w, v), representing the zoom, whirl, and vertex numbers. The paper aims to determine the energy E and angular momentum L corresponding to different periodic orbits.

In this reproduction, I reconstructed the numerical procedure using Python and reproduced both the benchmark values and the periodic-orbit branches.

2. Numerical Method

For a given eccentricity e and periodic orbit (z, w, v), the parameter

$$q = w + \frac{v}{z}$$

is first determined. Equation (3.12) is then solved numerically for the latus rectum λ. The corresponding energy and angular momentum are calculated using Eq. (3.14):

$$E = \sqrt{\frac{\lambda^2 - 4M\lambda + 4M^2 - 4M^2e^2}{\lambda(\lambda - 3M - Me^2)}}$$

$$L = \lambda\sqrt{\frac{M}{\lambda - 3M - Me^2}}.$$

The calculation was implemented in Python using SciPy, with the complete elliptic integral evaluated using `scipy.special.ellipk`.

3. Reproduction Results

I first reproduced the paper's example $(z, w, v) = (3, 0, 1)$ for $e = 0.2, 0.5, 0.8$.

There is an internal inconsistency in the paper: Eq. (3.12) defines

$$q = w + \frac{v}{z}$$

which gives $q = 1/3$ for $(3, 0, 1)$, while the text states $q = 1$. However, using $q = 1/3$ reproduces the numerical values reported in Eq. (3.15). Therefore, the numerical implementation follows the general definition $q = w + v/z$.

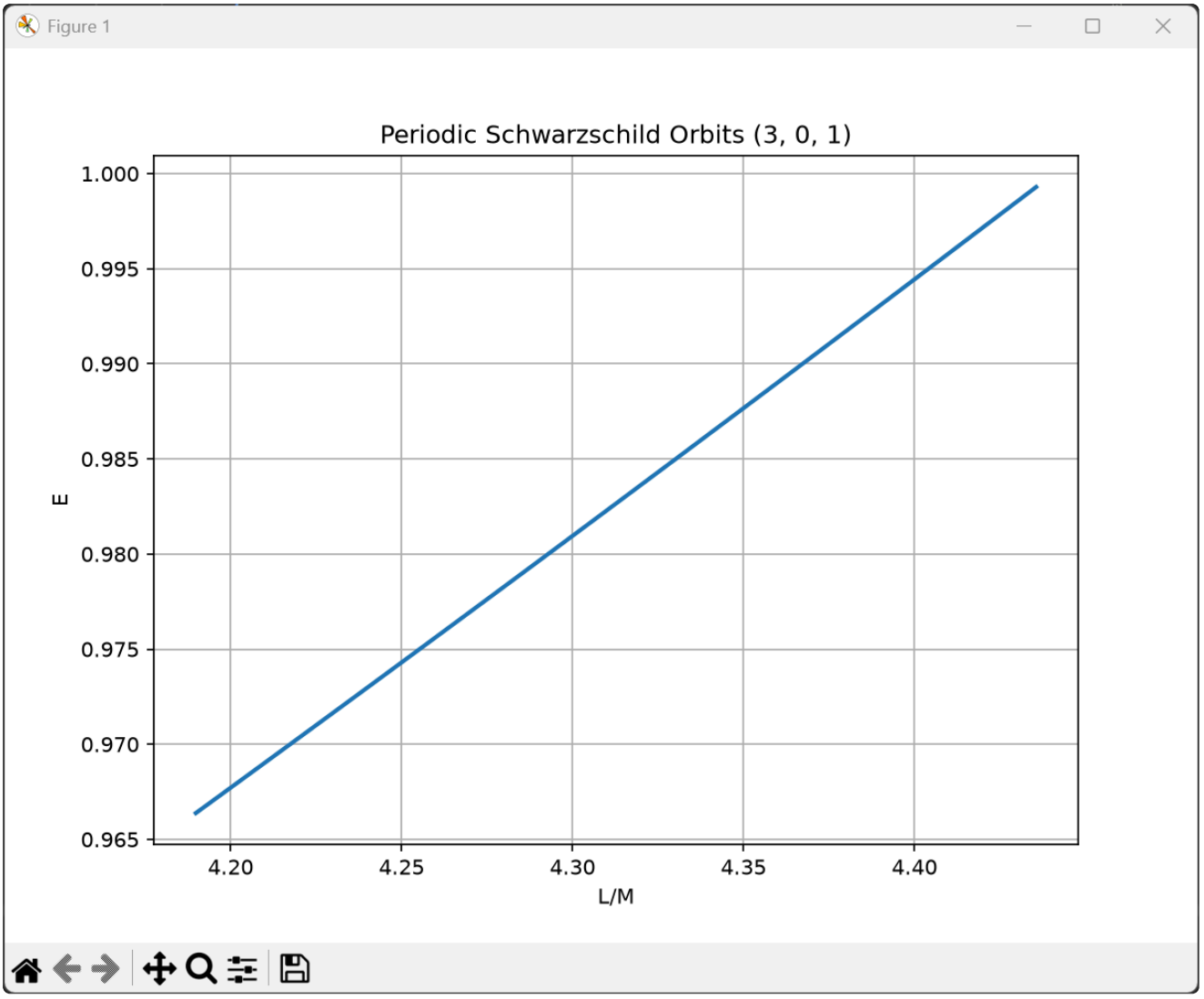
The reproduced values are:

e	λ/M	E	L/M
0.2	13.73203531	0.9676607919	4.199568981
0.5	13.82431376	0.9744720034	4.251258231
0.8	13.99183759	0.9875046766	4.348765910

These agree with the values given in Eq. (3.15) to numerical precision.

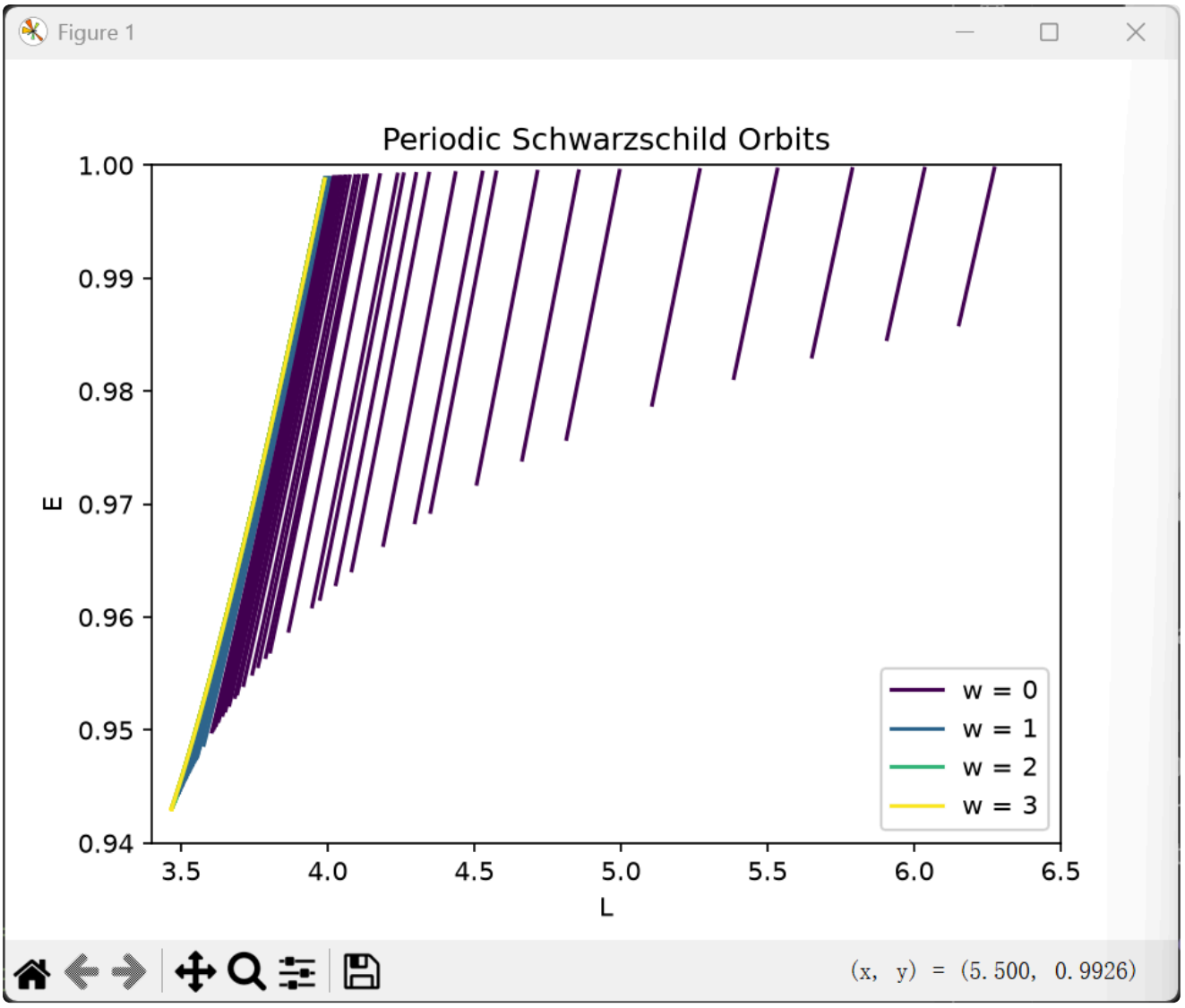
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e = 0.2, lambda/M = 13.73203531, E = 0.9676607919, L/M = 4.199568981
e = 0.5, lambda/M = 13.82431376, E = 0.9744720034, L/M = 4.251258231
e = 0.8, lambda/M = 13.99183759, E = 0.9875046766, L/M = 4.348765910
```

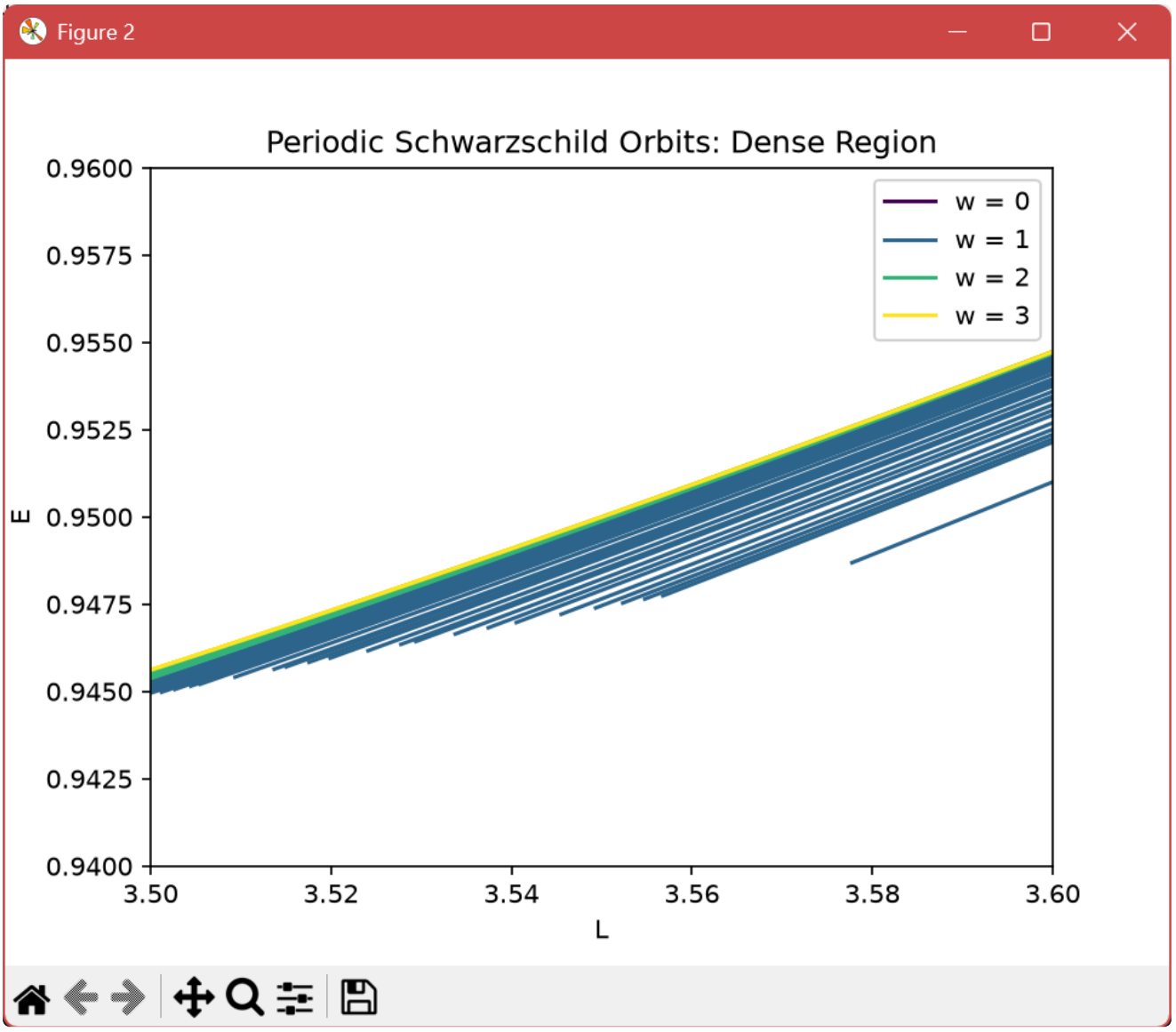
I then varied e from 0.01 to 0.99 to obtain the complete $(3, 0, 1)$ branch in the (L, E)-plane.



The resulting branch has positive slope and approaches $E = 1$ as $e \to 1$, consistent with the behavior described in the paper.

Finally, I extended the calculation to multiple allowed (z, w, v) combinations, scanning $1 \leq z \leq 10$, $0 \leq w \leq 3$, and $0.01 \leq e \leq 0.99$. The resulting branches reproduce the overall structure of the periodic-orbit distribution in the (L, E)-plane, including the dense region near the unstable circular-orbit branch.


Figure 1
Periodic Schwarzschild Orbits
1.00
0.99
0.98
0.97
0.96
0.95
0.94
E
3.5
4.0
4.5
5.0
5.5
6.0
6.5
L
w = 0
w = 1
w = 2
w = 3
(x, y) = (5.500, 0.9926)



4. Conclusion

The numerical reproduction successfully reconstructs the main procedure of the paper:

$(z, w, v), e \to q \to \lambda \to (L, E).$

The benchmark values and the resulting periodic-orbit branches are reproduced successfully. The calculation also revealed an internal inconsistency in the paper's statement of q for the $(3, 0, 1)$ example.

The next step is to reproduce the corresponding numerical results for the Melvin magnetic universe and investigate the effect of the magnetic-field-induced spacetime geometry on particle motion.